

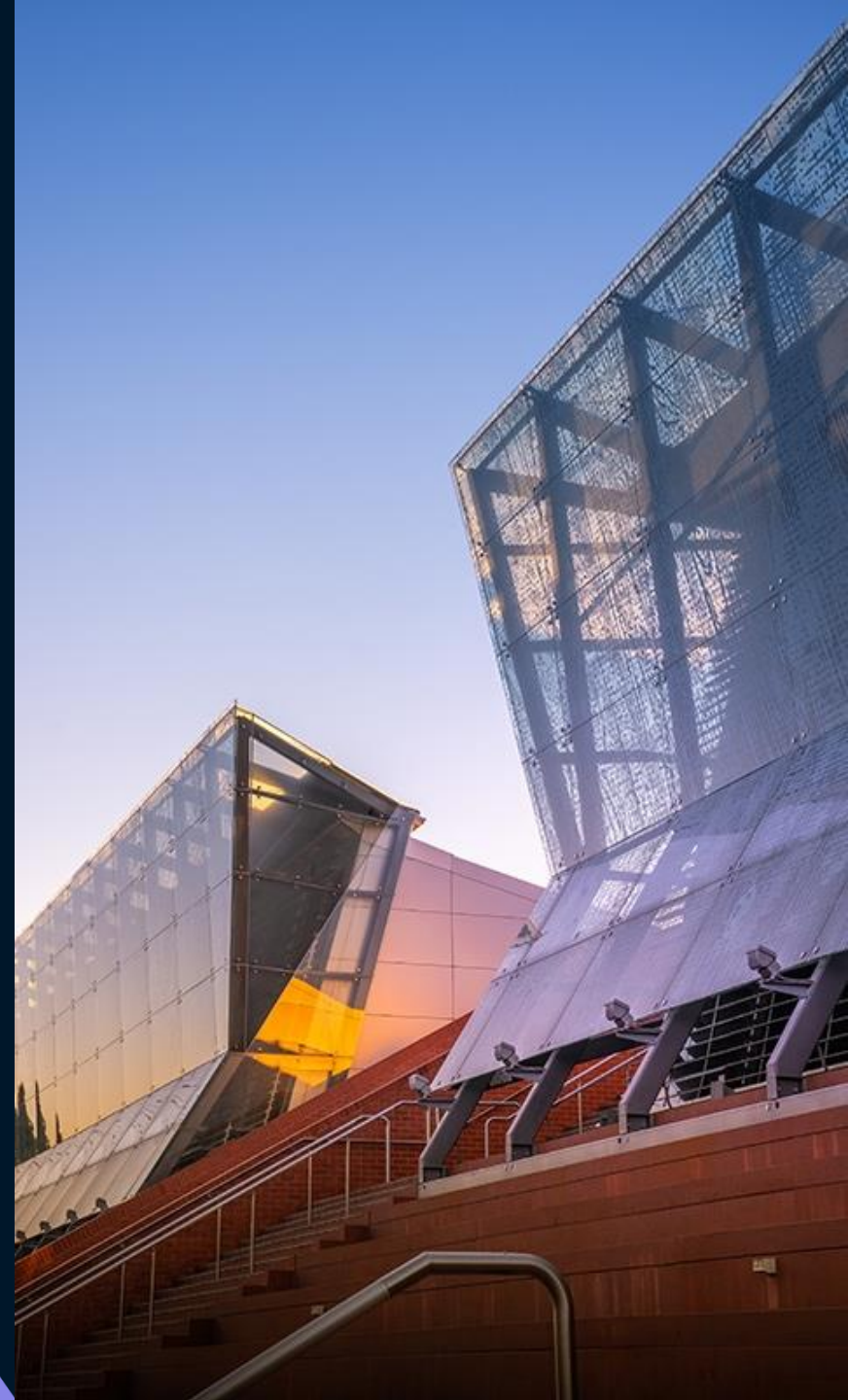


Edith Cowan University
School of Science, Center for Securing Digital Futures

Principal Supervisor Accreditation Program

AI in Academic Research

Dr. Ahmad Mohsin



Gen AI in Academic Research

A perspective for Supervisors and HDR Students

What is Gen AI: A Researcher Perspective?



Human- AI Collaboration

“Generative AI in Research utilises advanced models such as LLMs to generate content, significantly accelerating scientific discovery, enhancing creativity, and optimizing research workflows. It supports a wide range of applications, such as data generation, hypothesis development, literature review, and the production of novel research materials, streamlining complex processes across diverse academic disciplines”.

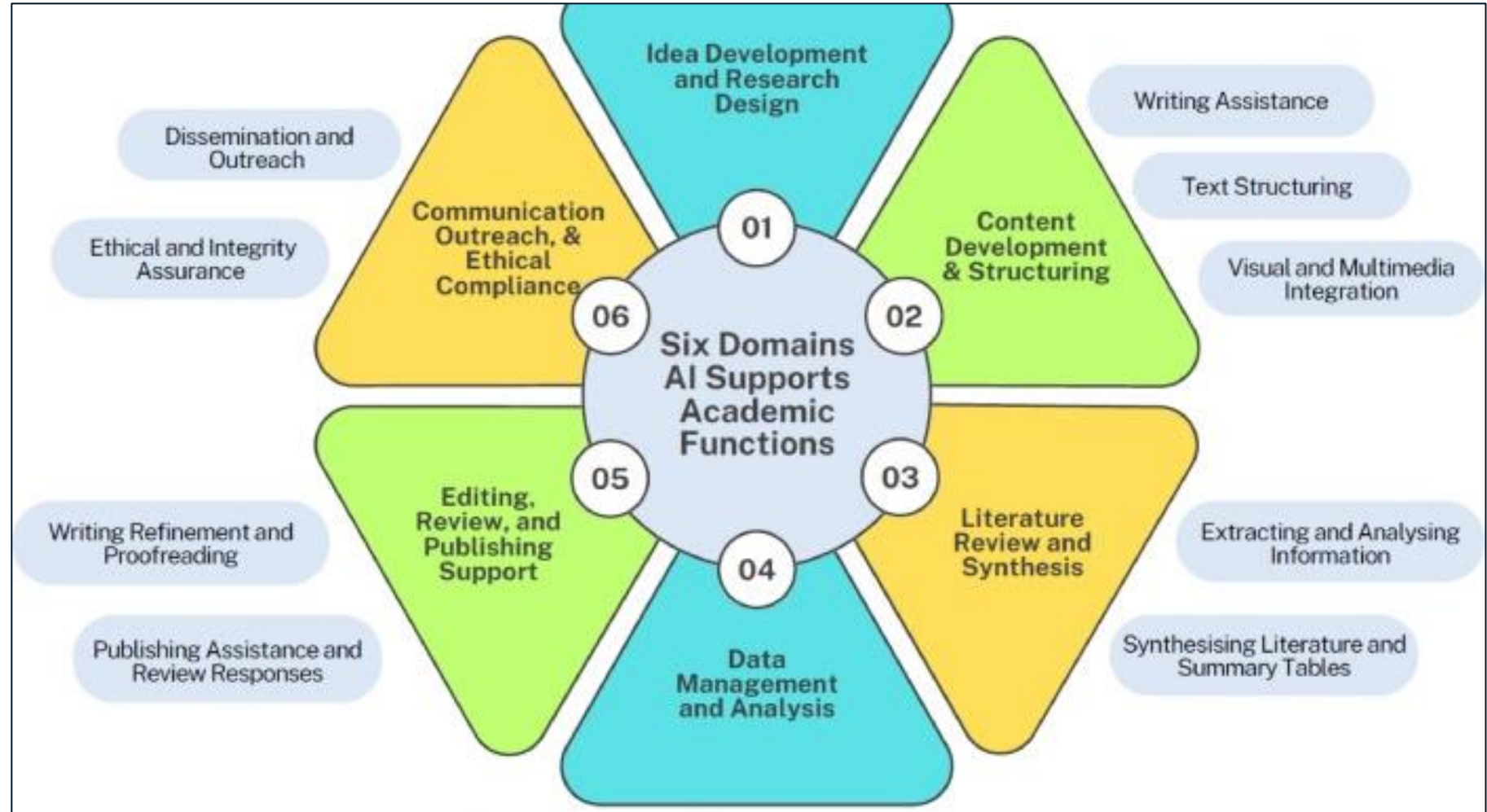
Use of Gen AI in Australian universities

A recent survey in September 2024 reveals that almost 71% of Australian university staff including Academics are using Generative AI Tools.

On Average 60% are using it for Literature search analysis Text Summary Analysis and day-to-day tasks.

10% use AI for Data Analysis, Visualisations and task Automation

How are Researchers using AI?



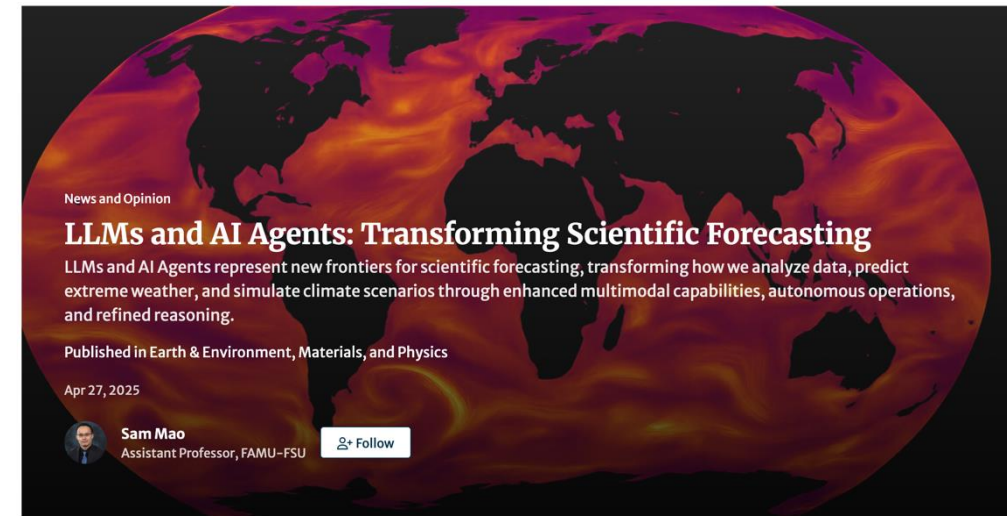
AI: Innovation and Discovery in Research

Artificial Intelligence, Scientific Discovery, and Product Innovation*

Aidan Toner-Rodgers[†]
MIT

November 6, 2024

This paper studies the impact of artificial intelligence on innovation, exploiting the randomized introduction of a new materials discovery technology to 1,018 scientists in the R&D lab of a large U.S. firm. AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings and a 17% rise in downstream product innovation. These compounds possess more novel chemical structures and lead to more radical inventions. However, the technology has strikingly disparate effects across the productivity distribution: while the bottom third of scientists see little benefit, the output of top researchers nearly doubles. Investigating the mechanisms behind these results, I show that AI automates 57% of “idea-generation” tasks, reallocating researchers to the new task of evaluating model-produced candidate materials. Top scientists leverage their domain knowledge to prioritize promising AI suggestions, while others waste significant resources testing false positives. Together, these findings demonstrate the potential of AI-augmented research and highlight the complementarity between algorithms and expertise in the innovative process. Survey evidence reveals that these gains come at a cost, however, as 82% of scientists report reduced satisfaction with their work due to decreased creativity and skill underutilization.



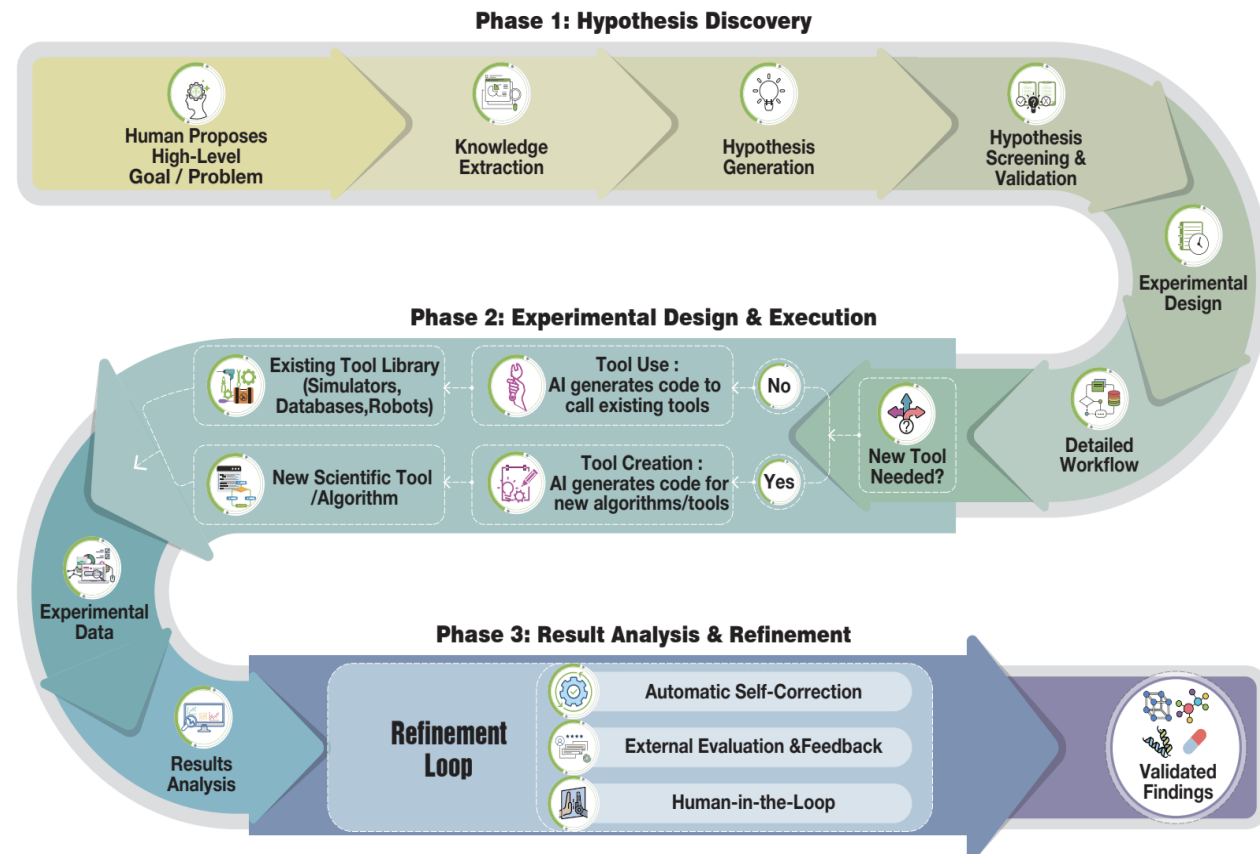
<https://communities.springernature.com/posts/llms-and-ai-agents-transforming-scientific-forecasting>

Autonomous Agents for Scientific Discovery: Orchestrating Scientists, Language, Code, and Physics

Abstract

Computing has long served as a cornerstone of scientific discovery. Recently, a paradigm shift has emerged with the rise of large language models (LLMs), introducing autonomous systems, referred to as agents, that accelerate discovery across varying levels of autonomy. These language agents provide a flexible and versatile framework that orchestrates interactions with human scientists, natural language, computer language and code, and physics. This paper presents our view and vision of LLM-based scientific agents and their growing role in transforming the scientific discovery lifecycle, from hypothesis discovery, experimental design and execution, to result analysis and refinement. We critically examine current methodologies, emphasizing key innovations, practical achievements, and outstanding limitations. Additionally, we identify open research challenges and outline promising directions for building more robust, generalizable, and adaptive scientific agents. Our analysis highlights the transformative potential of autonomous agents to accelerate scientific discovery across diverse domains.

<https://arxiv.org/pdf/2510.09901v2>



Researchers Concerns

“ GenAI is based, predominantly, on the unrecognised and unacknowledged work of the holders of Intellectual Property. It is hard to see how GenAI is ethical. **[M, Senior]**

“ GenAI... takes the work of scholars and writers and uses it to generate the lowest common denominator output without compensation to those it is literally stealing from. **[F, Academic]**

“ I consider generative AI to be a tool of plagiarism. The uses to date, especially in the creative industries (art, writing etc.) have involved machine learning that uses the creative works of others without permission. **[F, Academic]**

Human Intelligence and AI in Research

	Human Intelligence	AI Intelligence	HI+AI in Research
Creativity, Innovation	Human Intelligence is at top, where research needs innovation.	AI is an aiding tool for brainstorming of ideas around innovation but not actual innovation	Driven Through HI where philosophy is lead by Humans. AI can team up Humans to speed up innovation and creativity process.
Adaptability	HI can evolve according to complex situations and environments, can comprehend Philosophical thoughts	Good in known specific tasks for research, not able to adapt to new situations i.e. if a research hypothesis is failed?	HI, and human intellectual research though process to improve automation and aiding solving complex problems
Cognitive Processing	HI can process complex cognitive tasks required in a research	AI can do certain cogitative tasks at high speed i.e. Data Analysis, mathematical tasks in certain research stages	HI and AI can collaborate effectively for certain Research tasks.



AI and Research Journey stages

Early Candidature: Research Students

Develop Essential **Research methodologies**, **critical** thinking, formulating **research** ideas, research planning data collection, and Analysis.

Mid Candidature Stages

Brainstorming (Human collaborators Vs AI)
Learn to Develop Scientific AI Queries and **prompt Engineering**
AI-Driven Survey Existing methods

Any Stage

Editorial Assistance (Grammarly)
Proof Reading
Language Improvement

HDRS Maturity Stages

Scientific Research Methods, Domain Knowledge and Problem Rigor

AI- Coding
Data cleansing, Analysis, Visualizations

Towards Thesis Submission

Understand
AI policies prior to publication in Journals and Conferences
Avoid
Substantive use of AI (Thesis, journal articles)

Getting Ready for AI-Driven Research

- Fundamental Literacy for AI models and Tooling in Higher Education
- Universities must be at pace to respond how Students, Academics and Industry Partners are using AI in Research.
- Move from doom or hype to reality and should clearly know benefits and risks of AI driven Research.

Open Source and Commercial AI Tools for Academic Research

Leverage AI to push the frontier of scientific knowledge.

industry standards for the responsible use of AI in scientific discovery

Supervisors Role

- Supervisors must be equipped with necessary AI tooling and literacy
- Start open discussions among students concerning the use of AI.
- Encourage Research students to gain fundamental knowledge of AI tooling.
- Explore downstream AI models for Research purposes
- Provide explicit guidelines on using AI at different stages of a research journey.

Ethical Use of AI in Research (Students, Supervisors, and Org. Roles)

Research Supervisors and Students

- **Responsible** for Scientific Output regardless of AI usage
- **Language support is not a replacement** for HI critical thinking and contextual knowledge. Researchers should rely on critical thinking to supplement AI's **output**.
- **Transparency** and Integrity
- **Data** Privacy and Confidentiality

- Student proposal Development **(Not Desired)**
- Interpretation and Conclusions **(Don't)**
- AI-Peer Reviews and Assessments **(Don't)**
- Competitive Research Grants **(Subjective)**

- **Frameworks**, Guidelines Evolution
- Domain-Specific use of AI in Research
- AI **Governance**, Monitoring and Auditing
- **Leverage** AI to push the frontier of scientific knowledge

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Thank you.

